

Q What is a logistic function?
What is a logistic function's range of values?

Ans A logistic function is a type of sigmoid curve that models growth which starts exponentially but then levels off as it approaches a maximum limit.
The values of a logistic function will range from 0 to 1. The importance of x will vary from $-\infty$ to ∞ .

Q What makes logistic regression so well-liked?

Logistic regression is famous because it can convert the values of logits (log odds), which can range from $-\infty$ to ∞ to a range between 0 and 1. As logistic function outputs the probability of occurrence of an event, it can be applied to many real-life scenarios. It is for this reason that the logistic regression model is prevalent.

Q How can a logistic regression model's probability be stated as a conditional probability?

Ans $P(\text{discrete value of target variable } X_1, X_2, X_3 \dots X_n)$. it is the probability of the target variable taking up a discrete value (either 0 or 1 in case of binary classification problems) when the values of independent variables are given. For example, the probability an employee will resign (target variable) given his attributes such as his age, salary, KPIs, etc.

Q What are the outputs of the logistic model and the logistic function?

Ans The logistic model outputs the logit, which refers to the log odds, while the logistic function transforms this output into probabilities. The logistic model calculates a weighted sum of the input features, and this result (the logit) is then passed through the logistic function to obtain a probability value between 0 and 1.